Name _____ Student ID No. _____ Section _____

INSTRUCTIONS

1. You are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam room knowing that no additional time will be granted.
3. You must use a blue or black pen. Pencils are not allowed.
4. The use of any food or beverage is prohibited except in medical cases.
5. Electronic devices (e.g., cell phones) are prohibited.
6. All phones must be turned off and placed on the invigilator's desk in the exam room.
7. Books and reading materials must be placed on the invigilator's desk in the exam room.
8. You are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to attract attention of the proctor.
9. All the useful information required on this page before commencing your exam.
10. You are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. For student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the investigations office as per internal bylaws.

Read and abide with the exam policy. **Student's signature:** _____

No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CILO 1.3	14 (47%)			
2	Essay	CILO 2.5	16 (53%)			
Total			30 (100%)			

Review			
Reviewed by		**Signature & Date**	

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Answer	D	C	A	B	D	C	A

Q1 Event A satisfies $P(A) = P(\overline{A})$ then $P(A)$ equals

A) 0.25 B) 0.7 C) 0.4 D) 0.5 E) None

Q2 For the disjoint events A, B with $P(A) = P(B) = 0.1$ we have $P(A \cup B) =$

A) 0.1 B) 0.4 C) 0.2 D) 0.3 E) None

Q3 If X be a random variable with $Var(X) = 0.25$ and $E(X^2 - X) = 0$, then $E(X) =$

A) 0.5 B) 0.25 C) 0.75 D) 0.1 E) None

Q4 For independent events A, B with $P(A) = P(B)$ and $P(\bar{A} \cup \bar{B}) = 0.5$ then $P(A) =$

A) $\frac{1}{\sqrt{2}}$ B) $\sqrt{0.5}$ C) $\sqrt{0.4}$ D) 0.4 E) None

Q5 If $P(A \setminus B) = 0.1$ and $P(B) = 0.2$ then $P(A \cup B) =$

A) 0.1 B) 0.2 C) 0.4 D) 0.3 E) None

Q6 If $P(A) = 0.1$ and $P(\bar{A} \cap B) = P(\bar{B} \cap A)$, then $P(B) =$

A) 0.1 B) 0.2 C) 0.1 D) 0.4 E) None

Q7 If X be a random variable with $E(X) = E(X^3 + X)$ Then $E(X^3) =$

A) 0 B) 1 C) -1 D) -2 E) None

1) $p(\bar{A}) + p(A) = 1 \longrightarrow 2p(A) = 1$, $p(A) = 1/2$.

2) $P(A \cup B) = p(A) + p(B) - p(A \cap B)$ (0) disjoint

$= 0.2$.

3) $EX^2 - EX = 0 \rightarrow EX^2 = EX$

$0.25 = EX^2 - (EX)^2 \longrightarrow 0.25 = EX - (EX)^2$

$(EX)^2 - EX + 0.25 = 0$

by trying the choices:

A) $0.5 \longrightarrow (1/2)^2 - 1/2 + 1/4 = 1/4 - 1/2 + 1/4 = 1/2 - 1/2 = 0$.

4) $P(\bar{A} \cup \bar{B}) = \overline{P(A \cap B)} = 1 - p(A \cap B)$

$0.5 = 1 - p(A) \cdot p(B) = 1 - p^2(A)$

$p^2(A) = 1 - 0.5 = 0.5$, $p(A) = \sqrt{0.5}$

5) $P(A \cup B) = p(A/B) + p(B) = 0.2 + 0.1 = 0.3$

6) $p(A \setminus B) = p(B \setminus A)$

$p(A \cup B) - p(B) = p(A \cup B) - p(A) \rightarrow p(A) = p(B) = 0.1$.

7) $EX = EX^3 + EX \longrightarrow EX^3 = 0$.

Q8 (6 points): For the pdf $f(x) = e^{-x}$, $x > 0$ find the CDF and calculate $P(-1 < X < 1)$

$$P(-1<x<1) = P(0<x<1) = \int_0^1 e^{-x}dx = -e^{-x}\Big]_0^1 = 1-e^{-1} = 1-1/e$$

Q9 (5 points): For a discrete r.v x that takes values $x = 0, 5, 10$ each of equal probability, derive the CDF

$$\sum_{-\infty}^{\infty} p(x) = 1 \rightarrow p(0) = p(5) = p(10) = 1/3$$

$$F(x) = \begin{cases} 1, & x \geq 10 \\ 2/3, & 5 \leq x < 10 \\ 1/3, & 0 \leq x < 5 \\ 0, & x < 0 \end{cases}$$

Q10 (5 pts): A box contains 5 red balls, 4 blue balls, and 1 green balls. Three balls are drawn at random without replacement. What is the probability that we obtain at least one green ball ! (Leave your answer in terms of combinations; calculating the final numerical result is not required).

$$P = P(1G\&2B) + P(1G\&2R) + P(1G\&1B\&1R)$$

$$= \frac{\binom{1}{1}\binom{4}{2}\binom{5}{0}}{\binom{10}{3}} + \frac{\binom{1}{1}\binom{4}{0}\binom{5}{2}}{\binom{10}{3}} + \frac{\binom{1}{1}\binom{4}{1}\binom{5}{1}}{\binom{10}{3}}$$

$$= \frac{6}{120} + \frac{10}{120} + \frac{20}{120} = \frac{36}{120} = \frac{3}{10}.$$

OR P(1 Green & 2 not Green) → 9 balls

$$P = \frac{\binom{1}{1}\binom{9}{2}}{\binom{10}{3}} = \frac{3}{10}.$$

$\binom{n}{1} = n.$

$\binom{10}{3} = \frac{10!}{3!\,7!} = \frac{10\cdot 9\cdot 8\,7!}{3\cdot 2\cdot 7!} = 10\cdot 3\cdot 4 = 120.$

$\binom{5}{2} = \frac{5!}{2!\,3!} = \frac{5\cdot 4\cdot 3!}{2\cdot 3!} = 10.$

$\binom{4}{2} = \frac{4!}{2!\cdot 2!} = \frac{4\cdot 3\cdot 2}{2\cdot 2} = 6.$